

# Appendix to *The Wrong Razor*

## Complete Proofs

Michael Timothy Bennett

### A Complete proofs for *The Wrong Razor*

This appendix supplies complete proofs for the claims in the main draft that were previously given only as sketches. I keep the core Stack Theory definitions aligned with the master appendix. I only repeat the pieces needed to make the proofs readable and self-contained.

**Remark A.1** (Revision note). *This audit keeps the earlier repairs on intrinsic semantic length, the compatibility reading of exchangeable unseen requirements, the fixed-sample zero-empirical-risk PAC-Bayes scope, the nontriviality qualifier on finite recoding, the explicit weighted-forgetting bridge, and the finiteness assumption  $\mu(O_\alpha) < \infty$ . This version also removes the remaining scope drift between the manuscript and the appendix. In particular, the finite no-go result is stated only for nontrivial shorter-is-better selectors, the no-abstraction collapse claim is stated only for intrinsic semantic length with score symmetry at ties, and the Bayes-optimality theorem is stated as a compatibility theorem over unseen requirement sets rather than as an unrestricted parent-task claim. The underlying proofs are otherwise the same ones checked here. The main positive additions remain the direct prefix-free recoding proof, the literal lifted-policy version of the weighted weakness bridge, and the note that normalising PAC-Bayes on all of  $L_{\mathbf{v}}$  instead of only  $U$  changes the KL term by an additive constant only.*

#### A.1 Core finite definitions

**Definition A.2** (Environment, vocabulary, embodied language, extension, and weakness). *Let  $\Phi$  be a nonempty set of mutually exclusive states and let  $\mathcal{P} := 2^\Phi$  be the set of declarative programs. A finite vocabulary is a finite set  $\mathbf{v} \subseteq \mathcal{P}$ . Its embodied language is*

$$L_{\mathbf{v}} := \{l \subseteq \mathbf{v} \mid \bigcap_{p \in l} p \neq \emptyset\}.$$

*For  $l \in L_{\mathbf{v}}$  define the truth set*

$$T(l) := \bigcap_{p \in l} p, \quad T(\emptyset) := \Phi.$$

*A completion of  $x \in L_{\mathbf{v}}$  is any  $y \in L_{\mathbf{v}}$  with  $x \subseteq y$ . The extension of  $x$  is*

$$\text{Ext}(x) := \{y \in L_{\mathbf{v}} \mid x \subseteq y\}.$$

*The weakness of  $x$  is*

$$w(x) := |\text{Ext}(x)|.$$

**Interpretation.** A statement is a bundle of constraints a body can jointly realise. Its extension is the set of further commitments still compatible with it. Weakness counts how many such continuations remain open.

**Definition A.3** (Tasks and correct policies). A  $\mathbf{v}$ -task is a pair  $\alpha = \langle I_\alpha, O_\alpha \rangle$  with  $I_\alpha \subseteq L_{\mathbf{v}}$  and  $O_\alpha \subseteq \text{Ext}(I_\alpha)$ , where

$$\text{Ext}(I_\alpha) := \bigcup_{i \in I_\alpha} \text{Ext}(i).$$

A policy is a statement  $\pi \in L_{\mathbf{v}}$ . It is correct for  $\alpha$  when

$$\text{Ext}(I_\alpha) \cap \text{Ext}(\pi) = O_\alpha.$$

Write  $\Pi_\alpha$  for the set of correct policies.

**Interpretation.** A task fixes admissible inputs and the completions that count as correct. A correct policy is one whose completions on admissible inputs are exactly those correct outputs.

## A.2 Finite recoding no-go

**Definition A.4** (Occam prior on a finite hypothesis class). Let  $H$  be a finite hypothesis class. If  $\kappa$  is a prefix-free code on  $H$ , define the induced Occam prior by

$$P_\kappa(h) := \frac{2^{-|\kappa(h)|}}{\sum_{g \in H} 2^{-|\kappa(g)|}}.$$

**Proposition A.5** (Every strictly positive finite prior is an Occam prior within a factor of two). Let  $H$  be a finite hypothesis class and let  $Q$  be a strictly positive probability distribution on  $H$ . Then there exists a prefix-free code  $\kappa_Q$  on  $H$  such that the induced Occam prior satisfies

$$\frac{1}{2}Q(h) \leq P_{\kappa_Q}(h) \leq 2Q(h) \quad \text{for all } h \in H.$$

*Proof.* Enumerate  $H = \{h_1, \dots, h_n\}$  and set

$$\ell_i := \lceil -\log_2 Q(h_i) \rceil.$$

Then

$$\ell_i - 1 < -\log_2 Q(h_i) \leq \ell_i.$$

Exponentiating with base 2 gives

$$2^{-\ell_i} \leq Q(h_i) < 2^{-\ell_i+1} = 2 \cdot 2^{-\ell_i},$$

so equivalently

$$\frac{Q(h_i)}{2} < 2^{-\ell_i} \leq Q(h_i).$$

Also,

$$\sum_{i=1}^n 2^{-\ell_i} \leq \sum_{i=1}^n Q(h_i) = 1.$$

By the Kraft-McMillan theorem there exists a prefix-free code  $\kappa_Q$  with codeword lengths  $|\kappa_Q(h_i)| = \ell_i$  for all  $i$ . Let

$$Z := \sum_{j=1}^n 2^{-|\kappa_Q(h_j)|} = \sum_{j=1}^n 2^{-\ell_j}.$$

From the inequalities above,

$$\frac{1}{2} = \sum_{j=1}^n \frac{Q(h_j)}{2} < Z \leq 1.$$

Now

$$P_{\kappa_Q}(h_i) = \frac{2^{-\ell_i}}{Z}.$$

Using  $2^{-\ell_i} > Q(h_i)/2$  and  $Z \leq 1$  gives

$$P_{\kappa_Q}(h_i) \geq \frac{Q(h_i)/2}{1} = \frac{Q(h_i)}{2}.$$

Using  $2^{-\ell_i} \leq Q(h_i)$  and  $Z > 1/2$  gives

$$P_{\kappa_Q}(h_i) \leq \frac{Q(h_i)}{1/2} = 2Q(h_i).$$

Hence

$$\frac{1}{2}Q(h_i) \leq P_{\kappa_Q}(h_i) \leq 2Q(h_i) \quad \text{for all } i.$$

□

**Interpretation.** On a finite class, an Occam prior is only a recoding of an ordinary prior. The codebook carries the information.

**Lemma A.6** (Singleton semantic representation under no abstraction). *Assume  $\mathfrak{v} = \mathcal{P} = 2^\Phi$ . For every statement  $l \in L_{\mathcal{P}}$  the singleton  $\{T(l)\}$  lies in  $L_{\mathcal{P}}$  and satisfies*

$$T(\{T(l)\}) = T(l).$$

*Proof.* Because  $l \in L_{\mathcal{P}}$ , its truth set  $T(l)$  is nonempty. Since  $T(l) \subseteq \Phi$ , it is itself a program in  $\mathcal{P}$ . Therefore the singleton  $\{T(l)\}$  is a subset of  $\mathcal{P}$  with nonempty intersection, namely

$$\bigcap_{p \in \{T(l)\}} p = T(l) \neq \emptyset,$$

so  $\{T(l)\} \in L_{\mathcal{P}}$ . Its truth set is

$$T(\{T(l)\}) = \bigcap_{p \in \{T(l)\}} p = T(l).$$

□

**Corollary A.7** (No-abstraction collapse for intrinsic semantic length). *Fix a finite candidate family  $H \subseteq L_{\mathcal{P}}$  with  $\mathcal{P} = 2^\Phi$ . If  $\mathfrak{v} = \mathcal{P}$ , then every  $h \in H$  has an intrinsic semantics-preserving singleton representation of cardinality 1. Consequently any score function on  $H$  that depends only on that intrinsic semantic cardinality is constant on  $H$ . In particular, any probability distribution obtained by normalising such a score is uniform on  $H$ , and any selector that treats equal scores symmetrically cannot distinguish elements of  $H$ .*

*Proof.* By the lemma, every  $h \in H$  has the singleton representation  $\{T(h)\}$  and that representation has cardinality 1. So any score of the form

$$s(h) = g(|\{T(h)\}|)$$

for some function  $g$  is constant on  $H$ . Normalising a constant positive score gives the uniform distribution on  $H$ . More generally, if a selector treats equal scores symmetrically, then once every element has the same score it has no remaining information with which to rank one candidate above another.  $\square$

**Remark A.8** (Why the qualifier matters). *This corollary does not say that every arbitrary prefix-free code on  $H$  is uniform, nor that an external deterministic tie-breaker is impossible. An external codebook or tie-break rule can always be chosen after the fact and can reintroduce asymmetry. The point proved here is narrower and exactly the one needed for the no-go result. Once privileged encodings and hidden tie-breaks are stripped away, raw semantic length itself carries no discriminatory information.*

**Interpretation.** If abstraction is removed, intrinsic description length stops discriminating between candidates. The codebook was doing the work all along.

**Corollary A.9** (Finite recoding no-go for nontrivial shorter-is-better selectors). *Fix any selector that, whenever two candidates on a finite class have different description lengths, strictly prefers the shorter one. Then there exists a finite hypothesis class and two prefix-free codings of that same class on which the selector induces opposite rankings. Therefore no such nontrivial description-length selector defines a representation-invariant ideal of interactive intelligence on finite hypothesis classes.*

*Proof.* Let  $H = \{a, b\}$ . Define two prefix-free codings of the same class by

$$\kappa(a) = 0, \quad \kappa(b) = 10,$$

and

$$\kappa'(a) = 10, \quad \kappa'(b) = 0.$$

Both code sets  $\{0, 10\}$  and  $\{10, 0\}$  are prefix-free. The underlying hypothesis class has not changed, only which codeword is attached to which element. So this is a semantics-preserving recoding of the same finite class.

Under  $\kappa$  we have

$$|\kappa(a)| = 1 < 2 = |\kappa(b)|,$$

so any selector that strictly prefers the shorter description ranks  $a$  above  $b$ . Under  $\kappa'$  we have

$$|\kappa'(b)| = 1 < 2 = |\kappa'(a)|,$$

so the same selector ranks  $b$  above  $a$ . Therefore its ranking changes under a semantics-preserving recoding, which is exactly failure of representation invariance.  $\square$

**Remark A.10** (What the qualifier excludes). *A constant selector that ignores description length is of course representation invariant. It is also useless as a description-length principle. The no-go target is any selector that uses the shorter-is-better ordering in a nontrivial way.*

**Interpretation.** Once the selector uses the ordinary shorter-is-better ordering in any nontrivial way, a finite recoding can make it say opposite things on the same class. That is fatal if the ranking is meant to be invariant.

### A.3 Weakest-correct is Bayes optimal under exchangeable unseen requirements

**Definition A.11** (Exchangeable unseen-requirement model). *Fix a finite vocabulary  $\mathbf{v}$  and a child task  $\alpha$ . Write*

$$U := L_{\mathbf{v}} \setminus \text{Ext}(I_{\alpha})$$

*for the unseen region. For a correct child policy  $\pi \in \Pi_{\alpha}$  define its buffer by*

$$B_{\pi} := \text{Ext}(\pi) \cap U.$$

*A latent parent extension is a random set  $S \subseteq U$  of additional unseen requirements. The law of  $S$  is exchangeable if conditional on  $|S| = k$ , every  $k$ -subset of  $U$  is equally likely. A correct child policy  $\pi$  survives the extension  $S$  when*

$$S \subseteq B_{\pi}.$$

**Interpretation.** The child task fixes what has already been seen. The random set  $S$  is the unseen part of the future that may later matter. Exchangeability says that before those new requirements arrive, only their number matters, not their identities.

**Theorem A.12** (Weakest-correct is Bayes optimal for exchangeable unseen requirements). *Fix a finite vocabulary  $\mathbf{v}$  and a child task  $\alpha$ . Under the exchangeable unseen-requirement model, let*

$$\pi^{\star} \in \arg \max_{\pi \in \Pi_{\alpha}} w(\pi).$$

*Then*

$$P(S \subseteq B_{\pi^{\star}} \mid \alpha) = \max_{\pi \in \Pi_{\alpha}} P(S \subseteq B_{\pi} \mid \alpha).$$

*More generally, among all randomised selectors supported on  $\Pi_{\alpha}$ , the deterministic weakest-correct rule maximises expected survival probability.*

*Proof.* Fix  $\pi \in \Pi_{\alpha}$ . Because  $\pi$  is correct for the child task,

$$\text{Ext}(\pi) \cap \text{Ext}(I_{\alpha}) = O_{\alpha}.$$

Since  $L_{\mathbf{v}} = \text{Ext}(I_{\alpha}) \sqcup U$ , the extension of  $\pi$  splits as the disjoint union

$$\text{Ext}(\pi) = O_{\alpha} \sqcup B_{\pi}.$$

Hence

$$w(\pi) = |\text{Ext}(\pi)| = |O_{\alpha}| + |B_{\pi}|.$$

So on the fixed child task, maximising weakness is equivalent to maximising buffer size  $|B_{\pi}|$ .

Now condition on  $|S| = k$ . Under exchangeability, every  $k$ -subset of  $U$  is equally likely. The policy  $\pi$  survives exactly when every newly demanded unseen requirement lies in its buffer, that is,

$$S \subseteq B_{\pi}.$$

Therefore

$$P(S \subseteq B_{\pi} \mid \alpha, |S| = k) = \frac{\binom{|B_{\pi}|}{k}}{\binom{|U|}{k}},$$

with the convention that the numerator is 0 when  $k > |B_{\pi}|$ . For fixed  $k$  and  $|U|$ , this quantity is nondecreasing in  $|B_{\pi}|$ .

Let  $p_k := P(|S| = k \mid \alpha)$ . Averaging over  $k$  gives

$$P(S \subseteq B_\pi \mid \alpha) = \sum_{k=0}^{|U|} p_k \frac{\binom{|B_\pi|}{k}}{\binom{|U|}{k}}.$$

Each summand is nondecreasing in  $|B_\pi|$ , so the whole sum is nondecreasing in  $|B_\pi|$ . Because  $w(\pi) = |O_\alpha| + |B_\pi|$  and  $|O_\alpha|$  is fixed across  $\Pi_\alpha$ , every weakest correct policy maximises this survival probability.

Now let  $\Sigma$  be any randomised selector supported on  $\Pi_\alpha$ , with selection probabilities  $r_\Sigma(\pi)$ . Its expected survival probability is

$$\sum_{\pi \in \Pi_\alpha} r_\Sigma(\pi) P(S \subseteq B_\pi \mid \alpha).$$

This is a convex combination of deterministic survival probabilities, so it cannot exceed their maximum. Hence no randomised selector beats a weakest-correct deterministic selector.  $\square$

**Remark A.13** (Relation to parent-task language). *The counting argument here is a compatibility theorem over unseen requirements. If one specifies a parent-task construction in which correctness on the seen slice is held fixed and the only novelty is the unseen requirement set  $S$ , then the survival event  $S \subseteq B_\pi$  is exactly the event usually written as  $\pi \in \Pi_\omega$ . The proof itself does not need any stronger claim than compatibility with  $S$ .*

**Interpretation.** If you do not know which unseen constraints will arrive next, the safest correct move is to forbid as little of that unseen future as possible.

#### A.4 Regret is weighted forgetting

Fix a finite evaluation support  $\mathcal{X} = \{x_1, \dots, x_n\}$  with  $x_i = (h_i, T_i)$ , masses  $\mu_i > 0$  summing to 1, conditional probabilities  $p_i := p_{T_i}(h_i)$ , margins  $m_i := |p_i - \frac{1}{2}|$ , and optimal labels  $y_i \in \{0, 1\}$ . Fix a representation  $M$ . Write  $i \sim_M j$  iff  $M(h_i) = M(h_j)$  and  $T_i = T_j$ . Let  $\mathcal{C}_M$  be the set of resulting representation-test cells. An  $M$ -based policy chooses one report probability  $q_C \in [0, 1]$  for each cell  $C \in \mathcal{C}_M$ .

**Definition A.14** (Margin weight). *For each support point define*

$$\rho_i := \frac{2m_i}{\frac{1}{2} + m_i} \in [0, 1].$$

**Lemma A.15** (Pointwise regret decomposition). *Let  $\pi$  be an  $M$ -based policy with report probability  $q_C$  on cell  $C$ . For  $i \in C$ ,*

$$\delta_i(\pi) = \rho_i \left( (1 - q_C) \mathbf{1}\{y_i = 1\} + q_C \mathbf{1}\{y_i = 0\} \right).$$

*Proof.* If  $y_i = 1$ , then  $p_i \geq \frac{1}{2}$  and the optimal success probability is

$$V_i^* = p_i = \frac{1}{2} + m_i.$$

Under report  $q_C$ , the policy succeeds with probability

$$V_i^\pi = q_C p_i + (1 - q_C)(1 - p_i).$$

Hence

$$\delta_i(\pi) = 1 - \frac{V_i^\pi}{V_i^*} = 1 - \frac{q_C p_i + (1 - q_C)(1 - p_i)}{p_i}.$$

Since  $p_i - (1 - p_i) = 2m_i$ , this becomes

$$\delta_i(\pi) = \frac{2m_i}{\frac{1}{2} + m_i} (1 - q_C) = \rho_i (1 - q_C).$$

If  $y_i = 0$ , the same calculation with the roles of  $q_C$  and  $1 - q_C$  exchanged yields

$$\delta_i(\pi) = \rho_i q_C.$$

Combining the two cases gives the stated formula.  $\square$

**Definition A.16** (Margin-weighted forgetting cost). *Define*

$$K_\rho(M) := \min \left\{ \sum_{i \in B} \mu_i \rho_i \mid B \subseteq \{1, \dots, n\}, y \text{ is constant on } C \setminus B \text{ for every } C \in \mathcal{C}_M \right\}.$$

**Interpretation.** You delete just enough support points, weighted by their diagnostic importance, so that inside every cell the correct answer stops being ambiguous.

**Theorem A.17** (Exact reduction). *For every fixed representation  $M$ ,*

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^n \mu_i \delta_i(\pi) = K_\rho(M).$$

*Proof.* For each cell  $C \in \mathcal{C}_M$ , define

$$A_C := \sum_{i \in C, y_i=1} \mu_i \rho_i, \quad B_C := \sum_{i \in C, y_i=0} \mu_i \rho_i.$$

By the pointwise decomposition lemma, any  $M$ -based policy with report probability  $q_C$  contributes

$$A_C(1 - q_C) + B_C q_C$$

to expected regret on cell  $C$ . This is an affine function of  $q_C \in [0, 1]$ , so its minimum is attained at an endpoint and equals

$$\min\{A_C, B_C\}.$$

Summing over all cells gives

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^n \mu_i \delta_i(\pi) = \sum_{C \in \mathcal{C}_M} \min\{A_C, B_C\}.$$

Now compute the forgetting cost. To make the optimal label single-valued on a fixed cell  $C$ , one must delete all points labelled 1 in that cell or all points labelled 0 in that cell. Deleting the 1-labelled points costs  $A_C$  and deleting the 0-labelled points costs  $B_C$ . The minimum deletion cost on cell  $C$  is therefore  $\min\{A_C, B_C\}$ . Summing over cells yields

$$K_\rho(M) = \sum_{C \in \mathcal{C}_M} \min\{A_C, B_C\}.$$

Comparing the two displays proves the identity.  $\square$

**Interpretation.** The irreducible regret of a representation is exactly the weighted information it threw away.

**Definition A.18** (Lifted diagnostic task and weighted weakness). *Fix the representation  $M$  and introduce one unseen diagnostic output  $u_i$  for each support point  $x_i$ . Let*

$$U := \{u_1, \dots, u_n\}.$$

*Each  $u_i$  inherits the optimal label  $y_i \in \{0, 1\}$  and weight*

$$w_i := \mu_i \rho_i.$$

*For each representation-test cell  $C \in \mathcal{C}_M$ , an admissible lifted policy  $\vartheta$  chooses exactly one retained label  $\lambda_\vartheta(C) \in \{0, 1\}$ . Its lifted extension on  $U$  is*

$$\text{Ext}(\vartheta) \cap U := \{u_i \in U \mid y_i = \lambda_\vartheta(C_i)\},$$

*where  $C_i$  is the cell containing index  $i$ . Define the weighted weakness of  $\vartheta$  on the lifted task by*

$$W(\vartheta) := \sum_{u_i \in \text{Ext}(\vartheta) \cap U} w_i.$$

*Finally define*

$$W^*(M) := \max\{W(\vartheta) \mid \vartheta \text{ admissible on the lifted diagnostic task}\}.$$

**Interpretation.** The lifted task turns each diagnostic support point into one unseen output. An admissible lifted policy chooses, cell by cell, which label class to keep. Its weighted weakness is exactly the weighted unseen diagnostic future left open by that choice.

**Theorem A.19** (Exact Stack-Theoretic bridge). *Set*

$$w_i := \mu_i \rho_i \quad \text{and} \quad Z := \sum_{i=1}^n w_i.$$

*Then*

$$K_\rho(M) = Z - W^*(M).$$

*Equivalently,*

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^n \mu_i \delta_i(\pi) = Z - W^*(M).$$

*Proof.* Fix a cell  $C \in \mathcal{C}_M$ . Let

$$A_C := \sum_{i \in C, y_i=1} w_i \quad \text{and} \quad B_C := \sum_{i \in C, y_i=0} w_i.$$

An admissible lifted policy may retain only the label-1 outputs from  $C$  or only the label-0 outputs from  $C$ . Therefore the maximal retained mass on that cell is

$$\max\{A_C, B_C\}.$$



Summing over cells gives

$$W^*(M) = \sum_{C \in \mathcal{C}_M} \max\{A_C, B_C\}.$$

On the other hand,

$$Z = \sum_{i=1}^n w_i = \sum_{C \in \mathcal{C}_M} (A_C + B_C).$$

Hence

$$Z - W^*(M) = \sum_{C \in \mathcal{C}_M} (A_C + B_C - \max\{A_C, B_C\}) = \sum_{C \in \mathcal{C}_M} \min\{A_C, B_C\} = K_\rho(M),$$

where the last equality is the exact reduction theorem. Rearranging and substituting that theorem again yields

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^n \mu_i \delta_i(\pi) = Z - W^*(M).$$

□

**Interpretation.** This is the precise sense in which the regret floor is a weighted weakness deficit. The representation loses exactly the weighted diagnostic future that no admissible lifted policy can keep simultaneously open.

**Corollary A.20** (Policy-wise decomposition). *For each cell  $C \in \mathcal{C}_M$ , choose any*

$$q_C^* \in \arg \min_{q \in [0,1]} (A_C(1-q) + B_C q).$$

*Then every  $M$ -based policy  $\pi$  with report probabilities  $q_C$  satisfies*

$$\sum_{i=1}^n \mu_i \delta_i(\pi) = K_\rho(M) + \sum_{C \in \mathcal{C}_M} |A_C - B_C| |q_C - q_C^*|.$$

*In particular,*

$$K_\rho(M) \leq \sum_{i=1}^n \mu_i \delta_i(\pi),$$

*with equality exactly when the policy is cellwise optimal on every non-tied cell.*

*Proof.* Fix a cell  $C$ . If  $A_C > B_C$ , then the affine objective on  $C$  is minimised by  $q_C^* = 1$ . In that case

$$A_C(1 - q_C) + B_C q_C = B_C + (A_C - B_C)(1 - q_C) = \min\{A_C, B_C\} + |A_C - B_C| |q_C - q_C^*|.$$

If  $A_C < B_C$ , then  $q_C^* = 0$  and similarly

$$A_C(1 - q_C) + B_C q_C = A_C + (B_C - A_C)q_C = \min\{A_C, B_C\} + |A_C - B_C| |q_C - q_C^*|.$$

If  $A_C = B_C$ , then every  $q_C$  is optimal and both sides reduce to  $A_C$ . Summing over cells and using the exact reduction theorem gives the stated identity. The equality condition is immediate from the absolute-value term. □

**Interpretation.** This splits error into structural aliasing cost and avoidable within-cell misreporting.

**Corollary A.21** (Generalisation probability from regret). *Choose any admissible lifted policy  $\vartheta^\star$  with  $W(\vartheta^\star) = W^\star(M)$  and define its discarded set by*

$$D^\star := \{i \in \{1, \dots, n\} \mid u_i \notin \text{Ext}(\vartheta^\star) \cap U\}.$$

*Assume independent relevance events with probabilities*

$$r_i := 1 - e^{-\mu_i \rho_i}.$$

*Then*

$$K_\rho(M) = \sum_{i \in D^\star} \mu_i \rho_i,$$

*and the optimal  $M$ -based policy generalises with probability*

$$\Pr(\text{optimal } M\text{-based policy generalises}) = e^{-K_\rho(M)}.$$

*Proof.* By the exact Stack-Theoretic bridge,

$$K_\rho(M) = Z - W^\star(M) = \sum_{i \in D^\star} w_i = \sum_{i \in D^\star} \mu_i \rho_i.$$

Under the independent relevance model, the lifted policy generalises exactly when none of the discarded outputs become relevant. Therefore

$$\Pr(\text{generalises}) = \prod_{i \in D^\star} (1 - r_i).$$

Using  $1 - r_i = e^{-\mu_i \rho_i}$  gives

$$\Pr(\text{generalises}) = \prod_{i \in D^\star} e^{-\mu_i \rho_i} = \exp\left(-\sum_{i \in D^\star} \mu_i \rho_i\right) = e^{-K_\rho(M)}.$$

□

**Interpretation.** The regret floor is an exact odds penalty against future relevance.

## A.5 Continuous domains, invariance, and minimax optimality

From this point on, tasks still have finite input sets, so  $\text{Ext}(I_\alpha)$  is measurable as a finite union of measurable extensions. I also impose the standing condition

$$\mu(O_\alpha) < \infty,$$

where  $O_\alpha$  is the seen-correct region shared by all correct policies. Without that finiteness condition, every correct policy can have infinite  $\mu$ -weakness and the ordering claim becomes vacuous.

**Definition A.22** (Measured embodied language and continuous weakness). *A measured embodied language is a triple  $(\mathfrak{v}, \mathcal{A}_L, \mu)$  where  $\mathfrak{v} \subseteq \mathcal{P}$  is a vocabulary,  $\mathcal{A}_L$  is a  $\sigma$ -algebra on  $L_{\mathfrak{v}}$ , and  $\mu$  is a  $\sigma$ -finite measure on  $(L_{\mathfrak{v}}, \mathcal{A}_L)$  such that  $\text{Ext}(l) \in \mathcal{A}_L$  for every  $l \in L_{\mathfrak{v}}$ . The  $\mu$ -weakness of a statement  $l$  is*

$$w_\mu(l) := \mu(\text{Ext}(l)).$$

**Definition A.23** (Continuous extension model). *Fix a task  $\alpha$  and let*

$$U := L_{\mathbf{v}} \setminus \text{Ext}(I_\alpha)$$

*with  $0 < \mu(U) < \infty$ . A continuous extension model is a probability space carrying a random measurable subset  $S \subseteq U$  such that for all measurable  $A, B \subseteq U$  the events  $\{S \subseteq B\}$  and  $\{S \subseteq B, S \cap A \neq \emptyset\}$  are measurable. For a correct policy  $\pi \in \Pi_\alpha$  define its buffer by*

$$B_\pi := \text{Ext}(\pi) \cap U.$$

*Its generalisation score under the model  $P$  is*

$$G(\pi, P) := P(S \subseteq B_\pi).$$

**Definition A.24** ( $\mu$ -exchangeability and nondegeneracy). *Assume  $(U, \mathcal{A}_L|_U, \mu|_U)$  is an atomless standard finite measure space. A continuous extension model  $P$  is  $\mu$ -exchangeable if for every measure-preserving bijection  $\sigma : U \rightarrow U$  and every measurable  $B \subseteq U$ ,*

$$P(S \subseteq B) = P(S \subseteq \sigma(B)).$$

*It is nondegenerate if for every measurable  $A \subseteq U$  with  $\mu(A) > 0$ ,*

$$P(\emptyset \neq S \subseteq A) > 0.$$

**Interpretation.** Exchangeability says only the measure of unseen demands matters. Nondegeneracy says every positive-measure part of the unseen region can matter on its own.

**Lemma A.25** (Buffer measure determines  $\mu$ -weakness for correct policies). *Assume  $\mu(O_\alpha) < \infty$ . Let  $\pi \in \Pi_\alpha$  be correct and write*

$$O_\alpha := \text{Ext}(I_\alpha) \cap \text{Ext}(\pi).$$

*Then*

$$\text{Ext}(\pi) = O_\alpha \sqcup B_\pi \quad \text{and} \quad w_\mu(\pi) = \mu(O_\alpha) + \mu(B_\pi).$$

*Hence ranking correct policies by  $\mu$ -weakness is equivalent to ranking them by  $\mu(B_\pi)$ .*

*Proof.* Because  $L_{\mathbf{v}} = \text{Ext}(I_\alpha) \sqcup U$ , we have

$$\text{Ext}(\pi) = (\text{Ext}(\pi) \cap \text{Ext}(I_\alpha)) \sqcup (\text{Ext}(\pi) \cap U).$$

Since  $\pi$  is correct,

$$\text{Ext}(\pi) \cap \text{Ext}(I_\alpha) = O_\alpha.$$

By definition  $\text{Ext}(\pi) \cap U = B_\pi$ . Therefore

$$\text{Ext}(\pi) = O_\alpha \sqcup B_\pi.$$

Taking measures gives

$$w_\mu(\pi) = \mu(\text{Ext}(\pi)) = \mu(O_\alpha) + \mu(B_\pi).$$

The set  $O_\alpha$  is fixed across correct policies, so the two rankings agree.  $\square$

**Theorem A.26** (Exchangeable sufficiency). *If  $P$  is  $\mu$ -exchangeable, then there exists a nondecreasing function*

$$f_P : [0, \mu(U)] \rightarrow [0, 1]$$

*such that for every correct policy  $\pi \in \Pi_\alpha$ ,*

$$G(\pi, P) = f_P(\mu(B_\pi)).$$

*Equivalently, on correct policies,  $G(\pi, P)$  is a nondecreasing function of  $\mu$ -weakness.*

*Proof.* For measurable  $B \subseteq U$ , define

$$f_P(\mu(B)) := P(S \subseteq B).$$

We first show that this depends only on  $\mu(B)$ . Let  $B_1, B_2 \subseteq U$  be measurable with  $\mu(B_1) = \mu(B_2)$ . In an atomless standard finite measure space, equal-measure measurable sets are related by a measure-preserving bijection of the ambient space. So there exists a measure-preserving bijection  $\sigma : U \rightarrow U$  with  $\sigma(B_1) = B_2$ . By  $\mu$ -exchangeability,

$$P(S \subseteq B_1) = P(S \subseteq \sigma(B_1)) = P(S \subseteq B_2).$$

Thus  $f_P$  is well defined.

Now suppose  $\mu(B_1) \leq \mu(B_2)$ . Atomlessness gives a measurable subset  $B'_2 \subseteq B_2$  with  $\mu(B'_2) = \mu(B_1)$ . By the previous paragraph,

$$P(S \subseteq B_1) = P(S \subseteq B'_2).$$

Since  $B'_2 \subseteq B_2$ ,

$$\{S \subseteq B'_2\} \subseteq \{S \subseteq B_2\},$$

so

$$P(S \subseteq B_1) \leq P(S \subseteq B_2).$$

Hence  $f_P$  is nondecreasing. Applying this with  $B = B_\pi$  gives the stated formula. The equivalence with  $\mu$ -weakness follows from the lemma.  $\square$

**Interpretation.** Under exchangeable unseen growth, only buffer measure matters. Keeping more admissible future open cannot hurt.

**Theorem A.27** (Strict necessity). *If  $P$  is  $\mu$ -exchangeable and nondegenerate, and if two correct policies satisfy*

$$\mu(B_{\pi_1}) > \mu(B_{\pi_2}),$$

*then*

$$G(\pi_1, P) > G(\pi_2, P).$$

*Hence every maximiser of  $G(\cdot, P)$  on  $\Pi_\alpha$  is  $\mu$ -weakest.*

*Proof.* Because the space is atomless, there exists a measurable subset  $B'_1 \subseteq B_{\pi_1}$  with

$$\mu(B'_1) = \mu(B_{\pi_2}).$$

As above, equal-measure relabelling yields

$$P(S \subseteq B'_1) = P(S \subseteq B_{\pi_2}).$$

Let

$$R := B_{\pi_1} \setminus B'_1.$$

Then  $\mu(R) > 0$ . Since  $B_{\pi_1} = B'_1 \sqcup R$ ,

$$P(S \subseteq B_{\pi_1}) = P(S \subseteq B'_1) + P(S \subseteq B_{\pi_1}, S \cap R \neq \emptyset).$$

The second term is at least

$$P(\emptyset \neq S \subseteq R),$$

because every nonempty subset of  $R$  is contained in  $B_{\pi_1}$  and intersects  $R$ . By nondegeneracy this lower bound is strictly positive. Therefore

$$P(S \subseteq B_{\pi_1}) > P(S \subseteq B'_1) = P(S \subseteq B_{\pi_2}),$$

which is the claimed inequality.  $\square$

**Interpretation.** If one correct policy leaves strictly more positive-measure future open, a nondegenerate exchangeable world must sometimes reward that difference.

**Theorem A.28** (Unified optimality). *Let  $\mathcal{E}$  be the class of all  $\mu$ -exchangeable continuous extension models on  $U$ . For correct policies  $\pi_1, \pi_2 \in \Pi_\alpha$ ,*

1. *if  $\mu(B_{\pi_1}) \geq \mu(B_{\pi_2})$ , then  $G(\pi_1, P) \geq G(\pi_2, P)$  for every  $P \in \mathcal{E}$*
2. *if  $P \in \mathcal{E}$  is nondegenerate and  $\mu(B_{\pi_1}) > \mu(B_{\pi_2})$ , then  $G(\pi_1, P) > G(\pi_2, P)$*
3. *if  $\pi_1$  and  $\pi_2$  both maximise  $G(\cdot, P)$  on  $\Pi_\alpha$  for every nondegenerate  $P \in \mathcal{E}$ , then  $\mu(B_{\pi_1}) = \mu(B_{\pi_2})$ .*

*Consequently, among correct policies, maximising  $\mu$ -weakness is equivalent to maximising the generalisation score for every nondegenerate model in  $\mathcal{E}$ .*

*Proof.* Part 1 is the exchangeable sufficiency theorem. Part 2 is the strict necessity theorem. For Part 3, suppose both policies are optimal for every nondegenerate  $P \in \mathcal{E}$  but  $\mu(B_{\pi_1}) \neq \mu(B_{\pi_2})$ . Without loss of generality assume  $\mu(B_{\pi_1}) > \mu(B_{\pi_2})$ . Then Part 2 gives

$$G(\pi_1, P) > G(\pi_2, P)$$

for every nondegenerate  $P \in \mathcal{E}$ , which contradicts optimality of  $\pi_2$ . Hence the buffer measures must coincide.  $\square$

**Theorem A.29** (Minimax optimality of  $\mu$ -weakness). *Let  $\mathcal{E}$  be the class of all  $\mu$ -exchangeable continuous extension models on  $U$ . If a  $\mu$ -weakest correct policy  $\pi^* \in \Pi_\alpha$  exists, then*

$$\max_{\pi \in \Pi_\alpha} \inf_{P \in \mathcal{E}} G(\pi, P) = \inf_{P \in \mathcal{E}} \max_{\pi \in \Pi_\alpha} G(\pi, P),$$

*and both sides are attained by  $\pi^*$ .*

*Proof.* Let  $\pi^*$  maximise  $\mu$ -weakness on  $\Pi_\alpha$ . By unified optimality,

$$G(\pi^*, P) \geq G(\pi, P) \quad \text{for all } \pi \in \Pi_\alpha \text{ and all } P \in \mathcal{E}.$$

So for every  $P \in \mathcal{E}$ ,

$$\max_{\pi \in \Pi_\alpha} G(\pi, P) = G(\pi^*, P).$$

Taking the infimum over  $P$  gives

$$\inf_{P \in \mathcal{E}} \max_{\pi \in \Pi_\alpha} G(\pi, P) = \inf_{P \in \mathcal{E}} G(\pi^*, P).$$

Again by the same pointwise dominance,

$$\inf_{P \in \mathcal{E}} G(\pi^*, P) \geq \inf_{P \in \mathcal{E}} G(\pi, P) \quad \text{for all } \pi \in \Pi_\alpha.$$

Therefore  $\pi^*$  attains

$$\max_{\pi \in \Pi_\alpha} \inf_{P \in \mathcal{E}} G(\pi, P).$$

Combining the two displays yields the minimax equality, with both sides attained by  $\pi^*$ .  $\square$

**Interpretation.** The weakest correct policy is not merely good for one exchangeable model. It is pointwise optimal across the whole exchangeable uncertainty class, so it is automatically minimax.

**Definition A.30** (Policy map and function-preserving reparameterisation). *Let  $\Theta \subseteq \mathbb{R}^d$  be a parameter space and let  $F(\theta)$  denote the function computed by parameters  $\theta$ . For a vocabulary  $\mathfrak{v}$  over input-output behaviour, define the policy map*

$$\Pi_F(\theta) := \{p \in \mathfrak{v} \mid F(\theta) \text{ satisfies } p\}.$$

*A function-preserving reparameterisation is a diffeomorphism  $\phi : \Theta \rightarrow \Theta$  such that*

$$F(\phi(\theta)) = F(\theta) \quad \text{for all } \theta.$$

*Because the realised function itself satisfies every program in  $\Pi_F(\theta)$ , the policy map lands in  $L_{\mathfrak{v}}$ .*

**Theorem A.31** (Reparameterisation invariance of  $\mu$ -weakness). *If  $\phi$  is function-preserving, then*

$$\Pi_F(\phi(\theta)) = \Pi_F(\theta),$$

*and therefore*

$$w_\mu(\Pi_F(\phi(\theta))) = w_\mu(\Pi_F(\theta)).$$

*Proof.* If  $F(\phi(\theta)) = F(\theta)$ , then the two parameter values compute exactly the same input-output function. Therefore they satisfy exactly the same programs in the vocabulary  $\mathfrak{v}$ . That is,

$$\Pi_F(\phi(\theta)) = \Pi_F(\theta).$$

Applying the definition of  $\mu$ -weakness to these equal policies gives

$$w_\mu(\Pi_F(\phi(\theta))) = w_\mu(\Pi_F(\theta)).$$

$\square$

**Interpretation.** Weakness lives in function space, not coordinate space. If two parameter settings compute the same function, they leave exactly the same future open.

**Theorem A.32** (KL term is inverse buffer measure). *For a correct policy  $\pi$  with  $\mu(B_\pi) > 0$ , define the probability measures*

$$P_U := \frac{\mu|_U}{\mu(U)} \quad \text{and} \quad Q_\pi := \frac{\mu|_{B_\pi}}{\mu(B_\pi)}.$$

*Then*

$$\text{KL}(Q_\pi \| P_U) = \log \frac{\mu(U)}{\mu(B_\pi)}.$$

*In particular, the KL term is strictly decreasing in  $\mu(B_\pi)$ , and therefore strictly decreasing in  $\mu$ -weakness on correct policies.*

*Proof.* Relative to the base measure  $\mu|_U$ , the density of  $P_U$  is  $1/\mu(U)$  on  $U$ , while the density of  $Q_\pi$  is  $1/\mu(B_\pi)$  on  $B_\pi$  and 0 outside. Therefore the Radon-Nikodym derivative of  $Q_\pi$  with respect to  $P_U$  is constant on  $B_\pi$  and equals

$$\frac{dQ_\pi}{dP_U}(x) = \frac{\mu(U)}{\mu(B_\pi)} \quad \text{for } Q_\pi\text{-almost every } x.$$

So

$$\text{KL}(Q_\pi \| P_U) = \int_{B_\pi} \log \left( \frac{\mu(U)}{\mu(B_\pi)} \right) dQ_\pi = \log \frac{\mu(U)}{\mu(B_\pi)}.$$

This expression is strictly decreasing in  $\mu(B_\pi)$ . By the buffer-measure lemma, ordering by  $\mu(B_\pi)$  is the same as ordering by  $\mu$ -weakness on correct policies.  $\square$

**Corollary A.33** (Zero-empirical-risk PAC-Bayes ordering agrees with  $\mu$ -weakness). *Fix a sample size and a confidence level. Consider any PAC-Bayes upper bound whose zero-empirical-risk form is monotone increasing in the KL term. Then, among correct policies, that bound ranks policies exactly by decreasing buffer measure  $\mu(B_\pi)$ , and therefore exactly by decreasing  $\mu$ -weakness.*

*Proof.* At fixed sample size and confidence, the only policy-dependent quantity in such a zero-empirical-risk PAC-Bayes bound is the KL term. The theorem shows that this term is a strictly decreasing function of  $\mu(B_\pi)$ . So minimising the bound is equivalent to maximising  $\mu(B_\pi)$ , which is equivalent to maximising  $\mu$ -weakness on correct policies.  $\square$

**Remark A.34** (Normalising on  $U$  versus on all of  $L_v$ ). *Some presentations instead use the global prior*

$$P_0 := \frac{\mu}{\mu(L_v)}$$

*on all of  $L_v$ . For the same posterior  $Q_\pi$  one then gets*

$$\text{KL}(Q_\pi \| P_0) = \log \frac{\mu(L_v)}{\mu(B_\pi)} = \log \frac{\mu(L_v)}{\mu(U)} + \log \frac{\mu(U)}{\mu(B_\pi)}.$$

*So replacing  $P_U$  by  $P_0$  changes the KL term by an additive constant independent of  $\pi$ . The ordering by buffer measure, and therefore by  $\mu$ -weakness, is unchanged.*

**Interpretation.** In the PAC-Bayes story, the complexity term is again measuring how much future the policy rules out. The invariant object is weakness.